Lab Automation System — SRS samajh aur 5-day team plan

Project ka purpose

Electrical-appliance manufacturers ke lab mein switchgear, fuses, capacitors, resistors aur doosre products ki test records ko paper/Excel ke bajaye traceable application mein rakhna. Ye inventory/accounting application nahi; ye product identity, laboratory test records, department routing, results, remarks aur workflow status track karta hai.

SRS requirements

1. Product types/categories ke hisaab se modular configuration.

2. Har product family ke liye required/optional test types; har test department ke hisaab se route ho.

3. Har product ka unique 10-digit Product ID, product code/model, revision aur manufacturing number se linked.

4. Har test ka unique 12-digit Test ID, Product ID prefix, test code aur roll se linked.

5. Test record mein criteria, expected/actual output, result, detailed remarks, date aur tester(s) hon.

6. Product ID ya Test ID se lookup ho; advanced filters bhi hon.

7. Pending/in-progress/pass/fail aur product ka agla step dikhna chahiye.

8. Failure ke baad product re-manufacture/retest route mein jaye. **Current cycle ke tamam required lab tests PASS** hon tabhi CPRI readiness/handoff allow ho.

9. Approved role workflow: Lab Manager registers products; Administrator, Lab Manager, and Quality Control may assign pending tests to one or more available Testers; Testers cannot assign and complete only their own eligible current-cycle work. Lab Managers may edit instructions for an open pending test. QC confirms rework, can assign retests, records CPRI handoff/result, and records market release after approval. Rework—whether caused by a lab FAIL or CPRI rejection—requires all required tests again in a new cycle. Administrator keeps full access.

CPRI boundary

SRS CPRI API ya automatic data transfer specify nahi karta. App `CPRI Ready` ko gate karta hai, manually date/reference/user/notes ke saath handoff record karta hai, phir QC manually PASS/FAIL approval record karta hai. PASS ke baad final market release ek separate local action hai; FAIL product ko re-manufacture route mein bhejta hai. Koi CPRI API call nahi hoti; actual CPRI testing app ke bahar hoti hai.

ID split — sprint decision

User ne demo/eProject ke liye ye exact example split select kiya hai (final institutional rule samajh kar assume na karein; supervisor se verify karna ab bhi sensible hai):

- **Product ID (10 digits):** exact registered product code/model ka numeric mapping (2) + revision code (2) + manufacturing sequence (6).
- **Test ID (12 digits):** Product ID ke pehle 4 digits + test type numeric code (3) + per-product/per-test-type roll (5).

- Example: `SWG01` → model numeric code `01`; revision `01`; manufacturing `004417` → Product ID `0101004417`. Test type numeric code `001`, roll `00001` → Test ID `010100100001`.
- Mapping exact model/code ko identify karegi, sirf family ko nahi. Is liye `product_codes` registry use hoti hai. `product_types.numeric_code` family field Product ID generate karne ke liye use nahi hoti.
- Segments silently truncate nahi hote: invalid lengths reject hote hain; leading zeroes string ke taur par preserve hote hain; unique DB indexes + locked roll sequence collision rokta hai.
- Generated sequence reserve hone ke baad test insert fail ho to ek roll skip ho sakta hai, magar number reuse nahi hota.

Current repo/branch status

Baseline GitHub repo dark-teal PHP screens, product/test forms, advanced search aur test-type page rakhta tha. Full app Laravel mein rewrite nahi ki gayi; root-level pages ko URLs bachate hue incremental tareeqe se MVC mein migrate kiya ja raha hai.

Foundation pehle se implemented: MVC/config ka starting structure, env-driven `mysqli`, login/password verification/session hardening, page guards, selected role limits, CSRF, POST-only deletes, CSS-token dark/light theme, Vite build, migrations, demo seed.

Is sprint mein ab tak implemented:

- Product ID generator + exact product/model numeric registry; Add Product DB mapping se ID banata hai aur no-truncation validations karta hai.
- Test type par separate 3-digit numeric ID code; concurrency-safe per-product/per-test-type roll generator, legacy ID collision/backfill checks.
- Product families, exact product-code mappings, departments aur product-family test plan ke admin screens.
- Test entry: criteria/expected/actual/result/date; multi-tester participants; department ID; test-cycle assignment. The assignment form uses searchable tester checkboxes.
- PASS/FAIL/pending state updates; Admin/Manager/QC create pending assignments; Tester completes only their own eligible current-cycle pending test. One open test blocks additional assignments for a product; after closure only mapped test types not already assigned in the cycle remain eligible. Busy/inactive Testers are disabled and revalidated under transaction locks.
- Lab failure → re-manufacture → unused mapped tests are assigned in the next cycle; all required current-cycle tests must pass before `CPRI Ready`. Manager can edit instructions on open pending assignments; result correction is Administrator-only.
- QC-owned manual CPRI handoff, outcome/approval and market release; CPRI FAIL routes to re-manufacture; result, actor, time, references/notes, and release history stay attached to the original handoff with no API integration.
- Role-specific Administrator/Lab Manager/Tester/Quality Control dashboards and one route-role policy; forbidden links hide hote hain aur protected routes server-side 403 return karte hain. Tester dashboard/list/detail/search linked profile aur participant assignments se scoped hain.
- Shared authenticated shell/reusable cards/tables/forms ab dashboard, testing, product register, product catalogue, family plan, aur workflow views mein use ho raha hai. Public landing/About/Contact/Login/Register bhi shared public shell use karte hain; registration only Tester account + profile banata hai; contact message database mein save hota hai (email delivery nahi). Tailwind/Vite, locally bundled Inter + Font Awesome, modest reveal/hero motion, aur reduced-motion fallback included hain.
- Authenticated request par active state/role DB se refresh hota hai; 8-hour idle expiry, CSRF, password hashing, session regeneration retained hain.

- Earlier snapshots include historical import/migration/seed and HTTP smoke checks; they are not proof for the current tree. On 2026-10-12, JS syntax check, Vite build, and PHP grammar parsing passed. Native PHP, MySQL/MariaDB, and browser screenshot checks are unavailable here; current role gates, migration 010, busy/sequential assignments, preferences, and visual behavior still need local verification.

Abhi baqi / review: product-family test-plan sequence ko UI se custom reorder karna abhi available nahi; exact ID segment lengths abhi institutional confirmation maangte hain; several legacy CRUD screens still have separate content/layout shells. Migrations 009–010, role restrictions, sequential/busy-tester concurrency, profile preferences, CPRI result/release transitions, and the full regression flow need local MySQL/PHP verification; responsive, keyboard, contrast, and theme screenshots need a browser; teammate acceptance rerun and Aiven setup also remain. Production certification scope mein nahi.

5 din ke tasks — clear ownership

Member A — PHP/backend lead (project owner; strongest PHP)

- **Din 1:** Git branch/repo audit, config + shared DB baseline, migration review; instructor se ID split verify.
- **Din 2:** Auth/session/CSRF/role checks; Product/Test ID generation aur uniqueness ka ownership; code review.
- **Din 3:** Product/test persistence, product-family test-plan mapping, tester(s)/department data; failure/rework/retest transactions.
- **Din 4:** All-required-tests-pass → CPRI-ready/manual handoff gate, advanced search integration, DB migration/rollback checks.
- **Din 5:** End-to-end regression, access-control/security review, Aiven demo DB setup, release checklist (GitHub push only after team review).

Member B — Data, modules aur functional QA (basic/moderate PHP)

- **Din 1:** SRS traceability sheet, field dictionary, realistic samples aur 12–15 acceptance cases.
- **Din 2:** Product family, exact product-code, test-type, department aur family-test-plan screens verify; seed/demo data check.
- **Din 3:** Test-entry criteria/expected/actual/remarks/tester(s) fields, lists/detail screens aur input validation test.
- **Din 4:** PASS/FAIL/PENDING, different departments, duplicate IDs, lab FAIL→re-manufacture→full retest, CPRI PASS/FAIL→release/re-manufacture, incomplete CPRI gate chala kar bug list.
- **Din 5:** Full acceptance matrix rerun, demo reset/seed, QA sign-off aur reproducible bug notes.

Member C — Frontend aur documentation (basic/moderate PHP)

- **Din 1:** Existing pages/assets inventory, color-literal audit, theme toggle and dark/light baseline check.
- **Din 2:** Dark/light parity, responsive tables/forms, navigation and button/alert polish.
- **Din 3:** Advanced-search filters, validation/error/success states, status badges, workflow progress and detail-page UI.
- **Din 4:** Keyboard/form-label/contrast checks, Chrome/Edge/mobile visual testing; screenshots + UI bug list.
- **Din 5:** Setup/user guide, screenshots, demo checklist, final ZIP/repo check; `.env`, passwords, certificates share nahi karne.

Five-day acceptance gate

- Login ke baghair private pages nahi khulte; protected POST bina CSRF fail hota hai.
- Product save par mapping + revision + manufacturing sequence se unique 10-digit ID banta hai; wrong lengths truncate nahi hote.
- Test save par unique 12-digit ID; concurrent same product/test-type submissions ko alag roll milta hai.
- Criteria, expected/actual, result, remarks, date, department aur multiple testers save/retrieve hote hain.
- Test plan ke tamam required tests current cycle mein PASS hon tabhi CPRI handoff action chalta hai.
- Administrator, Lab Manager, and QC can create pending assignments; Tester cannot open `new-test.php`. One open current-cycle test blocks another assignment for that product; once closed, only unused mapped test types are eligible. Busy/inactive testers and unavailable products/tests are rejected server-side.
- Lab Managers can edit instructions on open pending assignments without changing actual output/result or Tester links. Result corrections are Administrator-only.
- Lab FAIL or CPRI rejection routes to re-manufacture; after QC records completion, an authorized assigner assigns eligible tests in a new cycle; previous tests and handoff/result history stay intact.
- Each role can edit their own profile and save theme/compact/reduced-motion preferences. Lab-wide Settings remain Administrator-only.
- CPRI PASS can be recorded only after a handoff; market release can be recorded only after PASS. Both are manual, actor/time-stamped, audited, and make no API call.
- Product ID aur Test ID se search, plus advanced filters, kaam karte hain.
- Har role ko sirf allowed tabs/actions dikhte hain; direct URL par bhi role guard 403 deta hai. Tester ki lists/details/search unke own assignments tak scoped hain.
- Public registration sirf Tester + linked profile banati hai; contact form message save karta hai aur email send hone ka claim nahi karta.
- Dashboard, testing page, product register/catalog, test-plan aur workflow ek shared responsive shell/components use karte hain.
- Dark/light choice pages ke darmiyan persist hoti hai; page CSS shared variables use karti hai.
- Shared DB par migrations one-time/idempotent hon; developers private `.env` se same remote DB ko use karen; credentials Git/chat mein nahi.

Five-day goal demonstrable eProject v1 hai. Aiven Free sirf demo/coursework ke liye; real laboratory ke liye backups/restore, high availability, TLS/network restrictions, security review aur institutional ID approval alag se lazmi hain.